

Project Briefing — English Translation

Cybersecurity and Social Engineering Project Consulting

Distributed Systems and Machine Learning

Class: 5th Semester, 2026

Theme: Distributed File System — HDFS and PySpark

Title: Using PySpark to Count Words

Opening Date: May 11, 2025

Closing Date: June 15, 2025

Complexity: HIGH

Integrative Project

Project 2 (May and June Term)

The Project

The second-term project will consist of two subprojects:

1. Process content from websites and social networks to count words and identify the occurrence of specific terms.
2. Present a dashboard that demonstrates the most frequent terms and highlights them in a positive and relevant way.

Project — Counting Words

Context

Your team belongs to a company that wants to carry out a marketing initiative across websites and social networks. To do this, it wants to identify the terms that should be associated with its product(s) and/or service(s), in a way similar to Google AdSense, in order to maximize the likelihood that the advertisement will appear positively to its target audience.

The strategy to be used is the Top of the Sales Funnel:

[Sales Funnel Reference](#)

“Top of the funnel is the first stage of the sales funnel, considered the attraction phase, when the visitor already knows they have a problem (or a need) and is looking for ways to solve it. In this first phase, it is necessary to spark people’s interest in order to guide them through the buying journey until they become customers.” (Nuvemshop)

Considering that Top of the Funnel is the stage in which it is necessary to attract and educate consumers, the team must create and distribute materials that are capable of achieving this on

the web. These materials may include, for example, specialized websites, blog posts, social media posts, e-books, videos, live streams, webinars, and similar content related to the business sector.

In this project, your team will not develop e-books, videos, or create websites, blogs, or social media channels for the market. Instead, the team must identify which are the best channels to advertise the product(s).

This project, which must be carried out in teams, consists of:

1. Identifying a business area, for example: Books, Perfumes, Sportswear, Food Products, Beverages.
2. Within the business area, identifying a specific Product/Service or a product line, for example: Books: a title or author; Sportswear: team jerseys; Beverages: energy drinks, juices, wines.
3. Identifying at least 20 websites (dynamic pages) and/or at least 1 social network with relevant content for promoting the product to its target audience.
4. Preparing the environment to process data from the selected websites and social networks (PySpark and Jupyter).

Dynamic pages are website pages whose content is generated dynamically from a data source. Examples include news websites, blogs, and web-based social networks, among others.

5. Processing this data (web scraping), counting words, and presenting the most frequent relevant words. Explain why the words are relevant for the product/service and target audience.
6. **Challenge:** Identify words/terms with high frequency but appearing in a negative context for the target audience (sentiment analysis).
7. Developing a dashboard to present the collected data, using tables and charts, in order to indicate which words/terms should be associated with the product/service being promoted.

Deliverables

1. A presentation (PowerPoint) with at least the following agenda:
 - Agenda
 - Group identification
 - Presentation of the business, product, and/or service
 - Identification of the target audience
 - Keywords searched
 - Websites and/or social networks analyzed
 - Research results
 - Word frequencies
 - Sentiment analysis (Challenge)

- Dashboards
- Analysis
- Conclusion (Decision-making)
- Selected websites and social network

2. Notebook (.ipynb)

3. Execution guide/manual for running the notebook

Other Contexts

As an example, this type of study may also be used to verify the **qualitative audience** of a product or service, such as in streaming platforms.

For example:

- Which character in a series is generating the most discussion on social networks?
- How is a new product/service being received in specialized digital media channels?
- How is the current marketing campaign for a product/service performing across digital media?

These are some examples of how the techniques required for this project can be applied.